Muhammad Mohsin

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Summary

Full-stack engineer with 6+ years building production systems across medtech and fintech, from embedded hardware to AWS. At Dosell I automated dispenser alarm processing with a cloud OCR pipeline (80% less manual work) and led the Flutter app behind a 1M SEK ARR milestone. Previously built a Bitcoin ATM network deployed across Spain, end to end from device protocols to backend. Works across Python/Django, TypeScript/Node, Flutter, AWS and Terraform, and owns systems from design to production.

Experience

Dosell AB Stockholm, Sweden

Full Stack Engineer *Dec 2022 - Present*

- Built a parallel OCR pipeline (Django, Celery) on AWS ECS with scheduled scaling, deployed as Terraform/Terragrunt IaC, auto-resolving about 680,000 dispenser alarms per year across 3,972 devices at 99.9%, cutting handling from 10+ minutes to 47 s per alarm (13x faster) and saving about USD 18,000 per year.
- Cut care-staff portal response times from 10-15 s to 1-3 s (5-10x faster) by adding read-replica routing, rewriting alarm queries, archiving and indexing audit events, and removing hidden outbound API calls in the NestJS/DocumentDB backend.
- Shipped Microsoft Entra ID SSO end to end (MSAL backend, group-to-role mapping, management UI), unlocking hospital-scale enterprise customers.
- Designed and built two products from scratch to production: WebLoad (auth, load/unload/merge flows, QR scanning, push notifications) and the Dosell Pro Flutter tablet app (patient UI, video calls, supervised dispense, multi-language).
- Engineered a production data lake (AWS DMS to S3 Parquet, Glue ETL decryption job, Athena) making 48.9 GiB queryable via SQL, plus daily Lambda FTP/SFTP crawlers archiving 365+ GiB of device files.
- Led development of Dosell's first hybrid application (Flutter, iOS/Android/Web) within a 2-month deadline, enabling a Finnish deal of 500-1000 units and securing the 1M SEK ARR milestone.
- Designed a microservices architecture with NestJS, PostgreSQL and event-driven patterns to replace legacy systems, increasing throughput 3x and reducing latency from 2000 ms to 300 ms.
- Architected React.js/Next.js medical applications with code splitting and responsive design, improving load times by 40% and achieving 99.9% uptime.
- Mentored five junior developers through code reviews, architecture guidance and workshops on React, Node.js and cloud deployment; all were promoted within 12 months.

Matrix Network International B.V. Remote

Software Engineer *Apr 2020 - Nov 2022*

- Architected a complete Bitcoin ATM solution fully remotely (hardware integration, AWS SQS microservices backend, website, Bitcoin mainnet/Lightning Network), deploying 22 ATMs across Spain for secure cash and debit-card purchases.
- Built the Venrecarga mobile recharge service with a multi-provider payment gateway (bunq, Stripe, PayPal, OpenNode), handling high-volume top-ups for the Venezuelan diaspora worldwide during the economic crisis.
- Developed the Rembit crypto remittance service over Mastercard and Visa networks with KYC and AML checks and a 24-hour automated S3-to-MySQL ETL pipeline via Athena, processing EUR 10,000+ in Bitcoin transfers in its first month.

Projects

CamBridge | *Creator (Swift, iOS + macOS)* | mohsin.se/cambridge-site

- Built two native Swift apps with zero third-party dependencies that turn an iPhone into a pro camera monitored and remote-controlled from a Mac or a second iPhone.
- Designed a custom wire protocol over Network.framework with hardware-encoded HEVC video (VideoToolbox, AVFoundation) and a SwiftUI interface.

DriveTest | *Full Stack Developer (Flutter + Django)* | mohsin.se/drivetest.html 2023 - Present

- Built and operate a cross-platform exam-prep app (Flutter + Django REST) helping aspiring taxi drivers and operators prepare for Swedish licensing exams, in 10+ languages.
- Implemented interactive practice tests, detailed explanations and progress tracking on iOS and Android.

Home Automation & Smart Vending | *Creator (Raspberry Pi, Python)*

- Built a full-stack home automation system with a web dashboard controlling lights and fans over MQTT, and a smart vending machine with bill-acceptor hardware, inventory tracking and automated dispense control.

Veg Shop Manager | *Flutter Developer*

- Built a Flutter app (Firebase, SQLite, Provider) for grocery store management in Spain with inventory, sales tracking, customer management and multi-language support.

Skills

Python: Django, Django REST Framework, Celery, pandas

JavaScript/TypeScript: Node.js, NestJS, Express.js, React.js, Next.js

Mobile: Flutter, Dart

AWS & Infrastructure: Terraform, S3, EC2, Lambda, RDS, DocumentDB, DynamoDB, SQS, Athena, CloudWatch, CloudFront, Docker, CI/CD, Linux

Databases: PostgreSQL, MySQL, SQL Server, Oracle DB, MongoDB, DynamoDB, Firestore

System Design: Microservices, event-driven architecture, Kafka, RabbitMQ, Elasticsearch/OpenSearch, Airbyte, ETL pipelines

Hardware: ESP32/ESP8266, Raspberry Pi, MDB protocol, bill & coin validators, serial communication, vending machines

AI Tools: Claude Code, GitHub Copilot, OpenAI, Gemini, Ollama, n8n

Awards & Achievements

– Oracle Certified Java Master Trainer

– Oracle Certified Java Foundations

Education

Blekinge Institute of Technology — Karlskrona, Sweden
MS Computer Science — *2022 - Present*

University of Gujrat — Gujrat, Pakistan
BS Computer Science — *2016 - 2020*